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1 library(dplyr)
2 student_data <- read.csv("C:/Users/admin/Downloads/student_data.csv")
3
4 # Show the details of students whose final_exam_mark is greater than 80%.
5 high_score <- student_data %>%
6   filter(final_exam_mark > 80)
7 high_score
8
9 # Arrange the dataset in ascending order based on course_work_mark
10 arranged_data <- student_data %>%
11   arrange(coursework_mark)
12 arranged_data
13
14 # Display only the student name and final_exam_mark attributes from the dataset
15 selected_data <- student_data %>%
16   select(student_name, final_exam_mark)
17 selected_data
18
19 # Specify the number of rows and columns in the dataset.
20 nrow(student_data)
21 ncol(student_data)
22
23 # Explain how to obtain the details of the given dataset using R.
24 # Show the dataset details based on the R query you execute.
25 str(student_data)
26 summary(student_data)
27 glimpse(student_data)
28
29 # How can you identify and detect outliers in the given dataset? Explain and demonstrate
30 # Calculate quartiles
31 Q1 <- quantile(student_data$final_exam_mark, 0.25)
32 Q3 <- quantile(student_data$final_exam_mark, 0.75)
33
34 # Calculate IQR
35 IQR_value <- IQR(student_data$final_exam_mark)
36
37 # Calculate boundaries
38 lower_limit <- Q1 - 1.5 * IQR_value
39 upper_limit <- Q3 + 1.5 * IQR_value
40
41 # Display values
42 Q1
43 Q3
44 IQR_value
45 lower_limit
46 upper_limit
47
48 # Detect outliers
49 outliers <- student_data[
50   student_data$final_exam_mark < lower_limit |
51   student_data$final_exam_mark > upper_limit,
52 ]
53
54 outliers
55 boxplot(student_data$final_exam_mark,
56   main = "Boxplot of Final Exam Marks",
57   ylab = "Final Exam Marks",
58   col = "lightblue")
59

```

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> library(dplyr)
> student_data <- read.csv("C:/Users/admin/Downloads/student_data.csv")
> # Show the details of students whose final_exam_mark is greater than 80%.
> high_score <- student_data %>%
+   filter(final_exam_mark > 80)
> high_score

```

	student_id	student_name	gender	coursework_mark	final_exam_mark
1	S007	David Jones	Male	64	84
2	S032	Joseph Smith	Male	94	82
3	S044	Jessica Brown	Female	77	84

```

> # Arrange the dataset in ascending order based on course_work_mark
> arranged_data <- student_data %>%
+   arrange(coursework_mark)
> arranged_data

```

	student_id	student_name	gender	coursework_mark
1	S028	Charles Garcia	Male	34
2	S001	Elizabeth Miller	Female	40
3	S006	Sarah Rodriguez	Female	41
4	S036	Karen Jones	Female	41
5	S038	Elizabeth Miller	Female	41
6	S037	David Davis	Male	42
7	S030	Barbara Williams	Female	43
8	S033	Richard Miller	Male	43
9	S040	Sarah Miller	Female	43
10	S043	Patricia Williams	Female	46
11	S013	John Smith	Male	50
12	S047	Linda Rodriguez	Female	50
13	S015	Elizabeth Garcia	Female	51
14	S022	Robert Miller	Male	51
15	S012	Mary Garcia	Female	52
16	S011	David Miller	Male	55
17	S019	Susan Williams	Female	55
18	S004	William Davis	Male	56
19	S039	Linda Williams	Female	57
20	S016	Robert Rodriguez	Male	58
21	S002	William Garcia	Male	59
22	S027	Charles Johnson	Male	59
23	S017	Linda Garcia	Female	61

24	S035	Barbara Miller	Female	61
25	S049	Linda Davis	Female	61
26	S048	Jessica Garcia	Female	63
27	S007	David Jones	Male	64
28	S023	David Martinez	Male	64
29	S021	Robert Miller	Male	66
30	S026	Charles Johnson	Male	66
31	S010	Jessica Garcia	Female	67
32	S014	John Brown	Male	67
33	S020	Joseph Rodriguez	Male	69
34	S009	Michael Garcia	Male	70
35	S046	Mary Davis	Female	70
36	S025	Joseph Williams	Male	72
37	S008	Thomas Martinez	Male	73
38	S050	Robert Brown	Male	73
39	S005	John Smith	Male	74
40	S041	Barbara Martinez	Female	75
41	S031	Joseph Williams	Male	76
42	S034	Richard Martinez	Male	76
43	S044	Jessica Brown	Female	77
44	S045	Susan Brown	Female	77
45	S003	Sarah Johnson	Female	79
46	S029	Linda Smith	Female	82
47	S024	William Jones	Male	87
48	S018	Sarah Rodriguez	Female	93
49	S032	Joseph Smith	Male	94
50	S042	Sarah Miller	Female	97
final_exam_mark				
1				39
2				41
3				35
4				45
5				34
6				38
7				55
8				42
9				51
10				38

10	38
11	53
12	54
13	51
14	54
15	55
16	49
17	44
18	62
19	70
20	55
21	63
22	48
23	74
24	54
25	53
26	54
27	84
28	55
29	62
30	57
31	65
32	58
33	62
34	64
35	60
36	73
37	60
38	57
39	56
40	55
41	66
42	67
43	84
44	77
45	64
46	68
47	64

```
48           70
49           82
50           73
> # Display only the student name and final_exam_mark attributes from the dataset
> selected_data <- student_data %>%
+   select(student_name, final_exam_mark)
> selected_data
```

	student_name	final_exam_mark
1	Elizabeth Miller	41
2	William Garcia	63
3	Sarah Johnson	64
4	William Davis	62
5	John Smith	56
6	Sarah Rodriguez	35
7	David Jones	84
8	Thomas Martinez	60
9	Michael Garcia	64
10	Jessica Garcia	65
11	David Miller	49
12	Mary Garcia	55
13	John Smith	53
14	John Brown	58
15	Elizabeth Garcia	51
16	Robert Rodriguez	55
17	Linda Garcia	74
18	Sarah Rodriguez	70
19	Susan Williams	44
20	Joseph Rodriguez	62
21	Robert Miller	62
22	Robert Miller	54
23	David Martinez	55
24	William Jones	64
25	Joseph Williams	73
26	Charles Johnson	57
27	Charles Johnson	48
28	Charles Garcia	39
29	Linda Smith	68
30	Barbara Williams	55

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29      Linda Smith      68
30 Barbara Williams    55
31   Joseph Williams    66
32      Joseph Smith    82
33   Richard Miller     42
34 Richard Martinez    67
35   Barbara Miller     54
36      Karen Jones     45
37      David Davis     38
38 Elizabeth Miller     34
39   Linda Williams     70
40   Sarah Miller       51
41 Barbara Martinez    55
42   Sarah Miller       73
43 Patricia Williams   38
44   Jessica Brown      84
45   Susan Brown        77
46   Mary Davis         60
47 Linda Rodriguez     54
48   Jessica Garcia     54
49   Linda Davis        53
50   Robert Brown      57
> # Specify the number of rows and columns in the dataset.
> nrow(student_data)
[1] 50
> ncol(student_data)
[1] 5
> # Explain how to obtain the details of the given dataset using R.
> # Show the dataset details based on the R query you execute.
> str(student_data)
'data.frame': 50 obs. of 5 variables:
 $ student_id      : chr  "s001" "s002" "s003" "s004" ...
 $ student_name    : chr  "Elizabeth Miller" "William Garcia" "Sarah Johnson" "William Davis"
 ...
 $ gender          : chr  "Female" "Male" "Female" "Male" ...
 $ coursework_mark: int   40 59 79 56 74 41 64 73 70 67 ...
 $ final_exam_mark: int  41 63 64 62 56 35 84 60 64 65 ...
> summary(student_data)

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> # Explain how to obtain the details of the given dataset using R.
> # Show the dataset details based on the R query you execute.
> str(student_data)
'data.frame': 50 obs. of 5 variables:
 $ student_id      : chr  "S001" "S002" "S003" "S004" ...
 $ student_name    : chr  "Elizabeth Miller" "William Garcia" "Sarah Johnson" "William Davis"
 ...
 $ gender          : chr  "Female" "Male" "Female" "Male" ...
 $ coursework_mark : int   40 59 79 56 74 41 64 73 70 67 ...
 $ final_exam_mark: int   41 63 64 62 56 35 84 60 64 65 ...
> summary(student_data)
      student_id      student_name      gender      coursework_mark
Length   :50   Length   :50   Length   :50   Min.    :34.00
N.unique :50   N.unique :42   N.unique : 2   1st Qu.:51.00
N.blank  : 0   N.blank  : 0   N.blank  : 0   Median  :62.00
Min.nchar: 4   Min.nchar:10   Min.nchar: 4   Mean    :62.42
Max.nchar: 4   Max.nchar:17   Max.nchar: 6   3rd Qu.:73.00
                                     Max.    :97.00

final_exam_mark
Min.    :34.00
1st Qu.:51.50
Median  :56.50
Mean    :57.88
3rd Qu.:64.75
Max.    :84.00
> glimpse(student_data)
Rows: 50
Columns: 5
 $ student_id      <chr> "S001", "S002", "S003", "S004", "S005", "S0...
 $ student_name    <chr> "Elizabeth Miller", "William Garcia", "Sara...
 $ gender          <chr> "Female", "Male", "Female", "Male", "Male",...
 $ coursework_mark <int> 40, 59, 79, 56, 74, 41, 64, 73, 70, 67, 55,...
 $ final_exam_mark <int> 41, 63, 64, 62, 56, 35, 84, 60, 64, 65, 49,...
> # How can you identify and detect outliers in the given dataset? Explain and demonstrate using R.
> # Calculate quartiles
> Q1 <- quantile(student_data$final_exam_mark, 0.25)
> Q3 <- quantile(student_data$final_exam_mark, 0.75)

```

```

fig R.
> # Calculate quartiles
> Q1 <- quantile(student_data$final_exam_mark, 0.25)
> Q3 <- quantile(student_data$final_exam_mark, 0.75)
> # Calculate IQR
> IQR_value <- IQR(student_data$final_exam_mark)
> # Calculate boundaries
> lower_limit <- Q1 - 1.5 * IQR_value
> upper_limit <- Q3 + 1.5 * IQR_value
> # Display values
> Q1
25%
51.5
> Q3
75%
64.75
> IQR_value
[1] 13.25
> lower_limit
25%
31.625
> upper_limit
75%
84.625
> # Detect outliers
> outliers <- student_data[
+   student_data$final_exam_mark < lower_limit |
+   student_data$final_exam_mark > upper_limit,
+ ]
> outliers
[1] student_id      student_name      gender            coursework_mark
[5] final_exam_mark
<0 rows> (or 0-length row.names)
> boxplot(student_data$final_exam_mark,
+   main = "Boxplot of Final Exam Marks",
+   ylab = "Final Exam Marks",
+   col = "lightblue")
> |

```